Postdoctoral Fellow, Columbia University | georgiannalin@gmail.com | www.bluelin.me

BLUE (GEORGIANNA) LIN

I am an interdisciplinary end-to-end developer, researcher, and designer. My work combines methods from human-computer interaction, machine learning, and applied AI to support user interpretation and decision-making with complex, multimodal data in consumer-facing products. Ask me about health agents and personalized insights.

Education and Academia

2025 -
Postdoctoral Research Fellow, Columbia University
Co-Advisors: Xuhai "Orson" Xu and Noémie Elhadad
Focus: Designing and building longitudinal, personalized health agents integrated with wearable and electronic health record data.

2021-2025
PhD Computer Science, University of Toronto
Dissertation: "**Multimodal Tracking with Ubiquitous Devices to Foster Holistic Menstrual Health Sensemaking**"
Co-Advisors: Khai Truong and Alex Mariakakis

- Led a team of 10+ to execute 2 longitudinal studies with 50 participants, collecting 40+ multimodal signals; conducted 100+ semi-structured interviews, surveys, and qualitative analyses.
- Built statistical and machine learning models including linear mixed-effects models and transfer learning frameworks to predict menstrual cycle events from wearable and hormonal data.
- Developed and deployed a React-based wearable data collection app with personalized health dashboards.
- Released one of the first public multimodal menstrual health datasets combining physiological, hormonal, and self-reported measures.

2019-2020
MS Computer Science, Georgia Institute of Technology
Focus: *Assistive wearables, head-worn displays, health informatics*

2016-2019
BS Computer Science, Georgia Institute of Technology

Recent Industry Experience

Sept 2022– July 2025
Student Researcher (Designing and Evaluating AI Health Agents), Google

- *Built LLM-powered prototypes to help users interpret personal health data via memory capture, goal refinement, and reflection.*
- *Conducted cross-cultural user studies (U.S., Japan) to assess usability, trust, and perceptions of AI-supported goal setting.*

April 2022 – August 2022
Research Intern (*Smart AI cooking assistants*), Samsung Electronics

- Designed and led a user study to extract design requirements for task-adaptive, multimodal AI cooking assistant.
- Results were presented to Samsung directors to inform product.

Jan 2021 – August 2021
Full-Stack Engineer (*Electrophysiology dashboards Startup*), Biocircuit Technologies

- Collaborated with clients and clinicians to design and engineer a full-stack application for collecting and analyzing nerve signals through wearable electrophysiology devices.

May 2020 – August 2020
Research Engineer Intern (*Youth wearable experiences*), Samsung Electronics

- Built wearable experience with voice augmentation and gesture-detection using smartwatch sensors.
- Worked with an interdisciplinary Think Tank Team of bioengineers, physicists, business strategists, designers, and legal on productization.

Select Recognition

2026	**Bill Buxton Dissertation Award (one student in Canada annually), $3000**
2025, '24, '23	**Department of Computer Science Award - Wolfond Scholarship, $15,000**
2024	**Alumni Association Distinguished Graduate Scholar, $1000**
2023	**Google PhD Fellowship, $110,200**
2022	**Mitacs Accelerate Award, $20,000**
2021	**Wolfond Fellowship, $20,000**
2020	**GVU Distinguished Masters Student Award, $5,000**

Select Publications

Journal Publications

1. **Georgianna Lin,** Rencong Jiang, Noémie Elhadad, and Xuhai "Orson" Xu 2026. A longitudinal health agent framework. Nature Health. (In review)
2. **Georgianna Lin,** Poojita Garg, Jin Yi Li, Rilong Zhang, Khai N. Truong, and Alex Mariakakis 2026. Inferring menstrual cycle phase labels from unlabeled wearable datasets for transfer learning NPJ Women's Health. (In review)
3. **Georgianna Lin,** Minh Ngoc Le, Khai N. Truong, and Alex Mariakakis. 2025**.** The Cognitive Strategies Behind Multimodal Health Sensemaking: A Menstrual Health Tracking Case Study. Proc. ACM Interact. Mob. Wearable Ubiquitous Technology **(**IMWUT '25**)**
4. **Georgianna Lin,** Minh Ngoc Le, Pierre Lessard , Gillian Einstein, Khai N. Truong, and Alex Mariakakis. 2025. From fertility to overall health: Barriers and catalysts to a holistic menstrual health perspective. ACM Transacations on Computing for Healthcare. (ACM Health '25)
5. **Georgianna Lin,** Helen Li, Ken Christofferson, Khai Truong, and Alex Mariakakis. 2024. Understanding wrist skin temperature changes to hormone variations across the menstrual cycle. NPJ Women's Health.
6. **Georgianna Lin**, Brenna Li, Jin Yi Li, Chloe Zhao, Khai Truong, and Alexander Mariakakis. 2024. Users' Perspectives on Multimodal Menstrual Tracking Using Consumer Health Devices. Proc. ACM Interact. Mob. Wearable Ubiquitous Technology (IMWUT '24).
7. **Georgianna Lin**, Rumsha Siddiqui, Zixiong Lin, Joanna Blodgett, Shwetak Patel, Khai Truong, and Alex Mariakakis. 2023. Blood glucose variance collected with continuous glucose monitors across the menstrual cycle. NPJ Digital Medicine.

Conference Papers

8. **Georgianna Lin,** Brenna Li, Pierre Lessard, Minh Truong, Khai Truong, and Alex Mariakakis. 2024. Functional Design Requirements to Facilitate Menstrual Health Data Exploration. In Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems (CHI '24).
9. **Georgianna Lin,** Jin Yi Li, Afsaneh Fazly, Vladimir Pavlovic, and Khai Truong**.** 2023. Identifying Multimodal Context Awareness Requirements for Supporting User Interaction with Procedural Videos. In Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems (CHI '23).
10. **Georgianna Lin**, Elizabeth D Mynatt, and Neha Kumar. 2022. Investigating Culturally Responsive Design for Menstrual Tracking and Sharing Practices Among Individuals with Minimal Sexual Education. In Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems (CHI '22).

Skills

Research: Human-Computer Interaction (HCI), Applied AI, Machine Learning, Data Visualization

Domains: Health Agents, Health Sensing, Wearables, Large Language Models, Context-Aware Systems, Augmented Reality, Accessibility

Quantitative Methods: Statistics, Data Cleaning & Preprocessing, Survey Design, Usability Test

Qualitative Methods: Interviews, Thematic Analysis, Affinity Mapping, Rapid Prototyping

Programming & Tools: Python, Tensorflow, R, Git, React, PostgreSQL, Figma

Other: Project Management, Product Building and Deployment, Grant and Human Research Ethics Writing, Presenting to Executives and International Audiences